

AI Mock Oral Examination - Question Bank

Subject: 310253 - Artificial Intelligence (TE Computer Engineering, SPPU 2019 Pattern)

50 Questions with Answers - Based on Laboratory Practicals 1 to 6

Subject Code	310253
Subject Name	Artificial Intelligence
Class	Third Year - Computer Engineering (TE)
University	Savitribai Phule Pune University (SPPU)
Pattern	2019 Pattern
Semester	II
Examination Type	Mock Oral / Practical Viva Voce
Total Questions	50
Practicals Covered	Practical 1 to Practical 6 (DFS/BFS, A*, Greedy, N-Queens, Chatbot, Expert System)

Instructions: These 50 questions are arranged practical-wise. Students should be able to answer each question in 1 to 3 sentences. Examiners may ask follow-up questions for clarification.

Section A: General Introduction to Artificial Intelligence

Q1. What is Artificial Intelligence (AI)?

Ans: Artificial Intelligence is the branch of computer science that deals with building machines and software capable of performing tasks that normally require human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making.

Q2. What is the Turing Test?

Ans: The Turing Test, proposed by Alan Turing in 1950, is a test of a machine's ability to exhibit intelligent behavior. If a human evaluator cannot reliably distinguish the responses of a machine from those of a human, the machine is said to have passed the test.

Q3. Define an Intelligent Agent.

Ans: An intelligent agent is anything that perceives its environment through sensors and acts upon that environment through actuators to achieve its goals. Examples include software agents, robots, and chatbots.

Q4. What does PEAS stand for?

Ans: PEAS stands for Performance measure, Environment, Actuators, and Sensors. It is used to describe the task environment of an intelligent agent.

Q5. List any four applications of Artificial Intelligence.

Ans: 1) Healthcare - medical diagnosis, 2) Finance - fraud detection, 3) Robotics - self-driving cars, 4) Natural Language Processing - chatbots and virtual assistants like Siri, Alexa.

Section B: Practical 1 - DFS and BFS Algorithms

Q6. What is graph traversal?

Ans: Graph traversal is the process of visiting (checking and/or updating) each vertex in a graph systematically. The two most common methods are Depth First Search (DFS) and Breadth First Search (BFS).

Q7. What is Depth First Search (DFS)?

Ans: DFS is a graph traversal algorithm that starts at the root node and explores as far as possible along each branch before backtracking. It uses a Stack (LIFO) data structure for its implementation.

Q8. What is Breadth First Search (BFS)?

Ans: BFS is a graph traversal algorithm that starts at the root node and explores all the neighbor nodes at the present depth (level) before moving on to nodes at the next depth level. It uses a Queue (FIFO) data structure.

Q9. Which data structure is used in DFS and BFS?

Ans: DFS uses a Stack (LIFO - Last In First Out). BFS uses a Queue (FIFO - First In First Out).

Q10. What is the time and space complexity of DFS?

Ans: Time Complexity: $O(V + E)$, where V is the number of vertices and E is the number of edges. Space Complexity: $O(V)$ for the stack and visited set.

Q11. What is the time and space complexity of BFS?

Ans: Time Complexity: $O(V + E)$, where V is vertices and E is edges. Space Complexity: $O(V)$ for the queue and visited nodes.

Q12. What is the main difference between DFS and BFS?

Ans: DFS goes deep into a branch first using a stack, while BFS explores all neighbors at one level first using a queue. BFS guarantees the shortest path in unweighted graphs; DFS does not.

Section C: Practical 2 - A* Algorithm for 8-Puzzle Problem

Q13. What is the A* (A-Star) algorithm?

Ans: A* is an informed (heuristic) search algorithm used for path-finding and graph traversal. It is a best-first search that finds the lowest-cost path from a start node to a goal node by combining actual cost and estimated cost.

Q14. Write the evaluation function used in A* algorithm.

Ans: $f(n) = g(n) + h(n)$, where $g(n)$ is the actual cost from the start node to node n , and $h(n)$ is the estimated (heuristic) cost from n to the goal node.

Q15. What is a heuristic function?

Ans: A heuristic function $h(n)$ is a function that estimates the cost of the cheapest path from node n to the goal. It guides the search toward the goal more efficiently. Example: Manhattan distance in 8-puzzle.

Q16. What is the 8-Puzzle Problem?

Ans: The 8-puzzle is a 3x3 grid containing 8 numbered tiles (1-8) and one blank space. The goal is to slide tiles into the blank space until they reach a specified goal configuration.

Q17. What is Manhattan Distance heuristic?

Ans: Manhattan distance is the sum of the absolute horizontal and vertical distances of each tile from its goal position. It is widely used as the heuristic $h(n)$ in solving the 8-puzzle problem.

Q18. What is admissibility of a heuristic?

Ans: A heuristic is admissible if it never overestimates the actual cost to reach the goal. An admissible heuristic guarantees that A* will find the optimal (shortest) solution.

Q19. List one advantage and one disadvantage of A* algorithm.

Ans: Advantage: A* is complete and optimal (finds the best solution if heuristic is admissible).
Disadvantage: It requires a lot of memory because it stores all generated nodes.

Q20. Name two heuristic functions used in the 8-puzzle problem.

Ans: 1) h1: Number of misplaced tiles (Hamming distance). 2) h2: Sum of Manhattan distances of all tiles from their goal positions.

Section D: Practical 3 - Greedy Search Algorithm

Q21. What is the Greedy algorithm?

Ans: A greedy algorithm is an optimization technique that builds up a solution piece by piece, always choosing the locally optimal (best at the moment) choice with the hope that it will lead to a globally optimal solution. It is simple, fast, but does not always guarantee the optimal solution.

Q22. What is Selection Sort? What is its time complexity?

Ans: Selection Sort is a comparison-based sorting algorithm that repeatedly finds the smallest element from the unsorted part and places it at the beginning. Time Complexity: $O(n^2)$.

Q23. What is a Minimum Spanning Tree (MST)?

Ans: A Minimum Spanning Tree of a connected, weighted, undirected graph is a subset of edges that connects all vertices with the minimum total edge weight and contains no cycles.

Q24. What is Prim's algorithm?

Ans: Prim's algorithm is a greedy algorithm that finds the Minimum Spanning Tree of a graph. It starts from any vertex and repeatedly adds the cheapest edge that connects a vertex inside the tree to one outside.

Q25. What is Kruskal's algorithm?

Ans: Kruskal's algorithm is a greedy algorithm to find the MST. It sorts all edges by weight in ascending order and adds them to the MST one by one, skipping any edge that forms a cycle.

Q26. What is Dijkstra's algorithm used for?

Ans: Dijkstra's algorithm is a greedy algorithm used to find the shortest path from a single source vertex to all other vertices in a weighted graph with non-negative edge weights.

Q27. What is the Job Scheduling problem in greedy approach?

Ans: Given a set of jobs with deadlines and profits, the goal is to schedule the jobs on a single processor (one job per unit time) to maximize total profit, where each job earns profit only if completed by its deadline.

Section E: Practical 4 - N-Queens / Graph Coloring (CSP)

Q28. What is a Constraint Satisfaction Problem (CSP)?

Ans: A CSP is a problem defined by a set of variables, each having a domain of possible values, and a set of constraints that specify allowable combinations of values. The goal is to find a value assignment that satisfies all constraints.

Q29. What are the three components of a CSP?

Ans: 1) X - a set of Variables. 2) D - a set of Domains for each variable. 3) C - a set of Constraints that must be satisfied.

Q30. What is the N-Queens problem?

Ans: The N-Queens problem is the task of placing N chess queens on an NxN chessboard so that no two queens attack each other - i.e., no two queens share the same row, column, or diagonal.

Q31. What is Backtracking?

Ans: Backtracking is a problem-solving technique that incrementally builds candidates for a solution and abandons (backtracks from) a candidate as soon as it determines the candidate cannot lead to a valid complete solution.

Q32. What is the Graph Coloring problem?

Ans: Graph coloring is the problem of assigning colors to the vertices of a graph such that no two adjacent vertices have the same color, using the minimum number of colors possible.

Q33. What is the Chromatic Number of a graph?

Ans: The chromatic number is the smallest number of colors needed to color the vertices of a graph such that no two adjacent vertices share the same color.

Q34. What are the types of constraints in CSP?

Ans: 1) Unary constraint - involves a single variable. 2) Binary constraint - involves two variables. 3) Global (higher-order) constraint - involves three or more variables.

Section F: Practical 5 - Elementary Chatbot using NLP

Q35. What is a Chatbot?

Ans: A chatbot is a software application that simulates human conversation through text or voice with users via natural language. It uses AI techniques like NLP to understand and respond to user queries.

Q36. What are the types of chatbots?

Ans: 1) Text-based chatbot - communicates via text interface. 2) Voice-based chatbot - communicates via human voice (e.g., Alexa, Siri, Google Assistant).

Q37. What is Natural Language Processing (NLP)?

Ans: NLP is a branch of AI that helps computers understand, interpret, and generate human (natural) languages such as English, Hindi, etc. It bridges the gap between human communication and computer understanding.

Q38. What are the two main approaches for building a chatbot?

Ans: 1) Rule-based approach - uses predefined rules and patterns. 2) Self-learning / AI-based approach - uses machine learning (Retrieval-based or Generative).

Q39. What is a rule-based chatbot?

Ans: A rule-based chatbot answers user queries based on a set of predefined rules and pattern matching. It is easy to build but cannot handle complex or unexpected queries.

Q40. What is NLTK?

Ans: NLTK (Natural Language Toolkit) is a popular Python library for working with human language data. It provides tools for tasks such as tokenization, stemming, lemmatization, parsing, and chatbot development.

Q41. List any two applications of NLP.

Ans: 1) Machine translation (e.g., Google Translate). 2) Sentiment analysis. 3) Chatbots and virtual assistants like Siri, Alexa. 4) Spam email detection. 5) Speech recognition.

Section G: Practical 6 - Expert System

Q42. What is an Expert System?

Ans: An Expert System is a computer application that solves complex problems in a specific domain at the level of a human expert. It uses a knowledge base of facts and rules along with an inference engine to draw conclusions.

Q43. What are the main components of an Expert System?

Ans: 1) Knowledge Base - stores facts and rules. 2) Inference Engine - applies rules to facts to derive conclusions. 3) User Interface - allows the user to communicate with the system.

Q44. What is a Knowledge Base?

Ans: The Knowledge Base is the component of an expert system that stores domain-specific facts and heuristic rules (usually in IF-THEN form) acquired from human experts.

Q45. What is an Inference Engine?

Ans: The Inference Engine is the brain of an expert system. It applies logical rules from the knowledge base to known facts to deduce new facts and make decisions or recommendations.

Q46. What is Forward Chaining?

Ans: Forward chaining is a data-driven inference method. It starts with the available facts and applies rules to derive new facts until the goal is reached. It answers - 'What can happen next?'

Q47. What is Backward Chaining?

Ans: Backward chaining is a goal-driven inference method. It starts with the goal (hypothesis) and works backward to find the facts that support the goal. It answers - 'Why did this happen?'

Q48. List any two examples of well-known Expert Systems.

Ans: 1) MYCIN - for diagnosing bacterial infections. 2) DENDRAL - for chemical analysis. Other examples include CADUCEUS, XCON, and PROSPECTOR.

Q49. What are the characteristics of an Expert System?

Ans: 1) High Performance, 2) Understandable (explains reasoning), 3) Reliable, 4) Highly Responsive, 5) Domain-specific expertise.

Q50. List any two applications of Expert Systems.

Ans: 1) Medical diagnosis (e.g., MYCIN). 2) Financial decision-making and stock trading. 3) Industrial process control. 4) Help-desk management. 5) Airline scheduling.

End of Question Bank - All the Best!